

Escaping Inequality in India: The Role of English-Medium Instruction (Selected Excerpts)

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What follows are excerpts from my 49-page independent research paper, “Escaping Inequality in India: The Role of English-Medium Instruction.” The paper’s key points are below.

1. **Question:** Does increasing baseline district-level EMI exposure (percentage of school-going children enrolled in EMI) affect consumption growth over the next decade?
2. **Data sources:** 2007-08/2017-18 microdata and metadata from the National Sample Survey, 35 datasets from the 2001 Census, and 2019-20 GIS shapefiles.
3. **Identification:** District-level 2SLS (pop.-weighted linguistic distance from Hindi IV, state-clustered SEs). Individual-level probit of education selection, design-based SEs.
 - **Key clarification:** State FEs, essential for 2SLS exclusion restriction, trigger extreme multicollinearity. District matching & possible FD-2SLS redesign needed.
4. **Main result:** Strong first-stage ($F = 39.2$), null second-stage estimate (0.2 pp, $p = 0.78$). Estimates are provisional pending district matching that enables state FEs.
5. **Next steps:**
 - Repair 2SLS exclusion restriction with district harmonization changes, FD-2SLS.
 - Redesign models for spatial spillovers, migration, ‘bad controls,’ heterogeneity.
 - Make response variable robust to local variation in inflation and shocks.

The next five pages include the following details:

- 6: **Research question**, contribution to literature, and summary/outline of paper.
- 23: **Instrumental variable**, relevance, issues with exclusion restriction ($\kappa = 28730.84$).
- 27: **Second-stage 2SLS results**; recall $F = 39.2$ in first stage.
- 28: **Interpretation of 2SLS**, small & noisy estimate, ‘bad control,’ spillover attenuation.
- 33: **District harmonization** assumptions, algorithm, and demonstration.

What is the effect of exposure to English-medium instruction enrollment in 2007-08 on the growth of local consumption from 2007-08 to 2017-18?

First, note the improvements made here over the previous literature. The setting of 2007-2018 may be more relevant today than the settings of similar literature, all of which are closer to the advent of service job offshoring in the 1990s (e.g., Refeque & Azad, 2022). Additionally, note that our response variable here is consumption, not wages. India holds a vast number of informal and self-employed individuals, meaning its labor force varies wildly in the degree to which wages reflect material well-being.¹¹

Second, we must highlight that unlike much of this literature, we lack household panel data. We will thus follow previous papers (e.g., Martin et al., 2017) and aggregate data at the district level.¹² More specifically, we will take cross-sectional data from 2007-08 (National Sample Survey Office, 2011) and 2017-18 (National Sample Survey Office, 2022), aggregate each at the district level separately, and match districts across the time periods to construct a district pseudo-panel. District-level variables are developed in Sec. 4.1 and the district matching method is explained in Sec. A.2 .

Our empirical methodology will then proceed in two parts. In Sec. 3, we study the composition of our key “EMI exposure” measure by estimating an individual-level probit of school participation in 2007-08. In Sec. 4, we move to the district level and implement two-stage least squares (2SLS): the weighted average of linguistic distance from Hindi in 2001 will be used to instrument for EMI exposure, onto whose fitted values we regress the growth rate of average consumption.¹³ The choice of instrument is discussed in Sec. 4.2.

Our 2SLS results will, at best, reflect the medium-run local general equilibrium effects of EMI exposure. On one hand, by aggregating away all individual or household-level variation, the ecological fallacy renders our 2SLS estimates uninterpretable at these lower levels. Additionally, school-going children in 2007-08 will still be quite young in 2017-18, and many may still be in school (particularly in more developed districts).¹⁴

With our current methodology, we find that greater exposure to EMI enrollment has a statistically and economically *insignificant* effect on local consumption growth (Sec. 4.4), in line with previous literature on EMI and Indian education quality. Note, however, that our outcome includes households not directly

¹¹See Shrivastav & Husain (2026) and Bahl & Sharma (2024) for the frequency and wage effects of informality respectively. Wages may still be more responsive to EMI in the medium-run window we study, however. See Sec. A.1 for more.

¹²A district in India is akin to a county in the United States, but with an average population of 2 million. Table 4 contains more information on district populations.

¹³Future work may incorporate both education selection and EMI exposure as endogenous regressors in a first-difference (FD) 2SLS design. The district-level aggregates of Table 1 may be able to function as education supply-shifting instruments.

¹⁴Cohort-specific measures of EMI exposure may thus be incorporated into future drafts, as may robustness checks that use school-reported data on medium of instruction to construct EMI exposure—with the aforementioned caveat that schools may lie about having EMI (Lahoti & Mukhopadhyay, 2019; Mathew & Srivastava, 2011). Such surveys may also reveal whether EMI exposure is rigid over time. If so, this could allow us to expand the window of time over which we can interpret our results. Rigidity could also rule out the agglomeration of EMI discussed in Sec. 4.4 as a potential attenuating force behind our results.

4.2 Instrumental Variable

Our instrument for $EMIE_d$ is $LingDistance_d$, the average linguistic distance from Hindi of the three most spoken mother tongues in district d in 2001, weighted by the number of speakers then. The mechanism behind this may seem odd: why choose linguistic distance from Hindi as opposed to, say, English? India’s large linguistic diversity²⁷ predisposes it to have large communication costs, large coordination costs, and thus hindered economic development within its borders (Nakagawa & Sugawara, 2021). To decrease such frictions in the market and in public goods provisioning (Liu & Pizzi, 2016), both Hindi and English have been thrust into the roles of competing domestic lingua franca (Clingsmith, 2014), meaning Hindi- and English-medium schools can be found across the country (Shastry, 2012, p. 294).

As the ‘distance’ between one’s mother tongue and Hindi increases, we expect the opportunity cost of learning English over Hindi to decrease. Because of this, we also expect individuals with a mother tongue further from Hindi to be more likely to enroll in EMI. “That distance from Hindi induces people to learn English” and even “schools to teach English” is in fact precisely what Shastry (2012, p. 289) claims. If we can then show that local linguistic distance induces exogenous variation in local EMI, we can wield it as an instrumental variable (IV) to isolate $EMIE_d$ ’s effect on consumption growth. Shastry helps us once more here, evidencing both the relevance (pp. 299-301) and exclusion restriction (pp. 302-305) of $LingDistance_d$ ’s effect on $EMIE_d$.

We are currently unable to replicate her justification of the exclusion restriction, however. Her argument centers on a map depicting the geographical balance of her residual variation, made possible despite the distinct regional divide in Figure 3 thanks to state fixed effects. In our case, adding state FEs explodes the condition number of our design matrix to $\kappa = 28730.84$ despite all individual collinearity measures remaining low (every scaled generalized variance inflation factor (GVIF) was below 6.05), with similar results for region FEs to a lesser degree. This almost certainly results from the many-to-many matching used in this paper’s district tracking algorithms.²⁸ Our plan to repair it moving forward is discussed in Sec. A.2.1.

A portion of this near-perfect multicollinearity may simply just be “God’s will” (Blanchard, 1987, p. 449): the States Reorganization Act of 1956, for example, aligned state lines with linguistic lines, a precedent which new states have followed since (Sarma, 2025). This possibility is precisely the motivation for our planned first-difference 2SLS design (Angrist & Pischke, 2008, p. 167). Even if only the data harmonization changes from Sec. A.2.1 are made, future work will still demean each district’s linguistic distance by subtracting away its state’s linguistic distance.²⁹

²⁷Perhaps the largest in the world, per one metric from Gurevich et al. (2025).

²⁸Many-to-many matching from 2001 to 2007-08 to 2017-18 was used to accurately reflect how real district changes occur as both mergers and partitions. While the intent was accuracy, the effect was degradation: the multiple duplicated rows for 2001 and 2007-08 measures in particular almost surely devastated the rank of the design matrix.

²⁹Dimensionality reduction (e.g., principal component analysis, ridge, lasso) and simultaneous equation estimation techniques

Table 6: Second-Stage Regression: Consumption Growth on EMIE (Fitted)

	Consumption Growth
	(1)
EMIE (fitted)	0.201 (0.710)
Consumption (2007-08)	-0.104*** (0.027)
Gini of Consumption (2007-08)	-11.654 (68.983)
Pct. Urban (ref: Rural)	1.240** (0.438)
Average HH size	-5.080 (6.090)
Dependency ratio $\times 100$	-0.419 (0.263)
Pct. Female head	-1.807 (2.167)
Pct. Hindu (ref: Other)	-0.404 (0.232)
Pct. Muslim	-0.786** (0.277)
Pct. Scheduled Tribe (ref: Other)	-0.857*** (0.238)
Pct. Scheduled Caste	-0.223 (0.293)
Pct. OBC	-0.615*** (0.177)
Pct. Small Land-owner (ref: No Land)	-0.169 (0.313)
Pct. Medium Land-owner	-0.717 (0.459)
Pct. Large Land-owner	0.567 (0.395)
Pct. Head Educ., Lit.-Primary (ref: Illiterate)	-0.384 (0.466)
Percentage. Head Educ., Secondary+	-0.494* (0.251)
Intercept	457.536*** (86.523)
Observations	454
R^2	0.232
Adjusted R^2	0.202

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered by state in parentheses.

instrumented by LingDistance_d with no covariates;³³ and finally 4) from estimating our current specification,

³³Which would allow us to obtain a LATE interpretation without nonparametric estimation or further parametric specification, per Blandhol et al. (2022). Conversely, see Chen (2022) for the conditions under which we can get a conditional LATE.

with both IV and covariates.³⁴

In the meantime, as would be expected in our first stage, greater consumption expenditures and urbanization associate with greater EMIE. After controlling for urban/rural settings, it seems that owning at least a small amount of land is associated with greater EMIE too. Also note the -40.45 estimate ($p = 0.043$) on the Gini coefficient: EMIE is lower in more unequal districts. At first, this seems to contradict our literature review in Sec. 1 and its implication that EMI exists within an inequality-amplifying positive assortative matching between households and schools (e.g., Muralidharan, 2019). Larger Gini coefficients, however, often imply longer right tails in income/wealth distributions (Benhabib & Bisin, 2018). We’d thus expect unequal districts to be characterized by a large proportion of poor and a small number of rich households—that is, by a large proportion of students in mother tongue-instruction and a small proportion in EMI. Also reassuring is the small but significant negative result on baseline consumption in the second stage, implying conditional beta-convergence (as described by Vu, 2013) between poorer and richer districts.

The consistency between these results and the literature makes our small and noisy estimate for EMI exposure’s effect on consumption growth all the more noteworthy. We find that an ostensibly exogenous percentage point increase in EMIE (i.e., in the percent of a district’s school-going children who are enrolled in EMI) associates, at the insignificant p -value of 0.78, with a 0.2 percentage point change in the growth rate of average household consumption expenditures from 2007-08 to 2017-18. Household-level decisions to bet on the human capital and English-language skills possible under EMI do not aggregate into local development when features of household context are accounted for.

This result may be driven by our large urbanization estimate in the second stage (1.24 at $p = 0.005$). Greater EMIE plausibly induces more urbanization: perhaps English skills lead to more service sector firms (Shastri, 2012) and rural-to-urban migration locally (see Sec. A.2.2), or perhaps migration to EMI causes EMI supply to agglomerate and thus even more EMI demanders to migrate. In either case, urbanization would be an outcome of, and thus “bad control” for, EMIE (Angrist & Pischke, 2008, pp. 47–51). Lagged urbanization (see Table 4 for the measure currently used) and the interaction effects of Sec. A.1 should address this.³⁵

Migration and spatial spillovers could also explain our small $\widehat{\text{EMIE}}$ coefficient. If EMI helps a person exit to distant urban IT hubs, for example, then the only effect of their EMI on the district they were educated in may be of remittances they send back. It was once widely accepted that migration in India only occurs over short distances, though some new evidence now contests this. Sec. A.2.2 discusses the debate in detail.

³⁴Multiple specifications could help us confirm relevance and covariate constructions (e.g., by testing the effect of EMI under different proxies of ability, as Azam et al. (2013) do), but they may be helpful in other ways as well. A positive OLS coefficient on EMIE_d despite a near-zero IV coefficient on EMIE_d , for example, could imply that while EMIE can *predict* local growth, any ostensible causality there is actually driven by unobserved features of households’ contexts which correlate with their EMIE.

³⁵What they may not address, however, is possible collider bias; see Schneider (2020) for an excellent review.

This seems to be precisely the same assumption made by India State Stories (2024) and Jaacks et al. (2020), whose data, despite containing some factual errors and typos, was the basis of our district matching method. To justify this assumption we turn to H. Kumar & Somanathan (2016), who up until 2001 are able to track the *proportion* of each district’s population that was allocated into a new district as a result of district changes. They label these changes as one of two types: clean partitions and non-partitions (which would include name changes, clean mergers, carve-outs, and border shifts). By extracting their results using Tabula (Aristarán et al., 2018), we know that 93.4% of non-partitions were either name changes or transfers of uninhabited land (H. Kumar & Somanathan, 2016). These district changes are plotted in Figure 4.

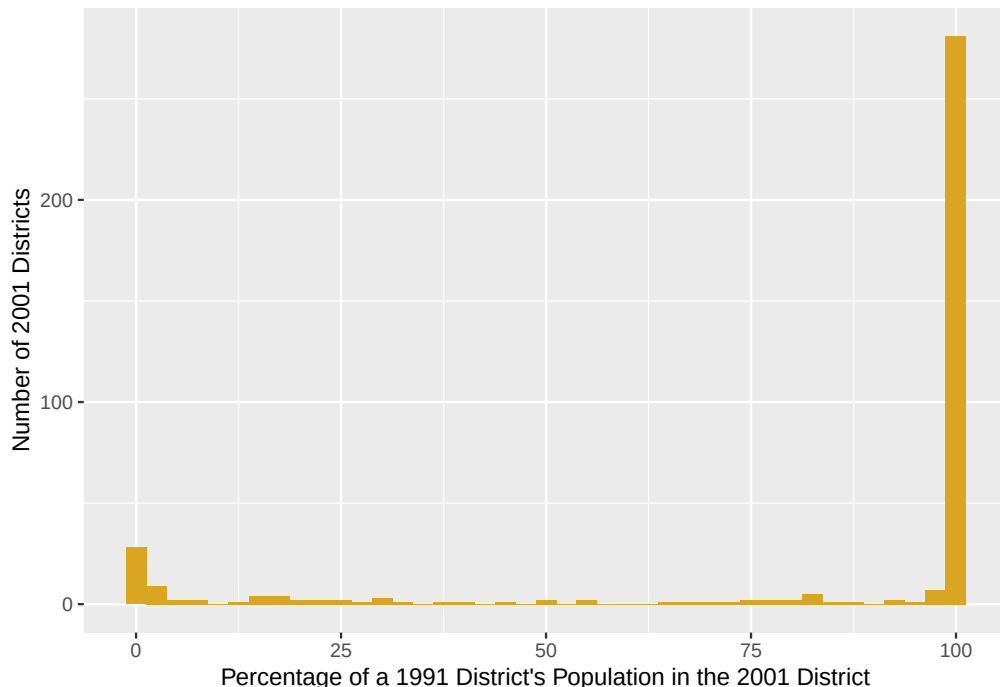


Figure 4: Number of 2001 districts which absorbed a percentage of a 1991 district’s population via name change, clean merger, carve-out, or border shift. Data from Kumar & Somanathan (2016).

We thus assume no district changes were carve-outs or border shifts. To match districts, we ran a hierarchical, cascading sequence of iterated fuzzy joins. Data from 2001, 2007-08, and 2017-18 is first linked to the closest year in the district trackers of India State Stories (2024) and Jaacks et al. (2020). Districts are then matched if they meet a certain string/phonetic distance threshold. Matches were made in the following order:

1. `soundex` = 0, for phonetically equivalent variations in anglicization (e.g., “Baleswar” and “Balasore”).
2. `qgram` distance = 0, for rearrangements of bigrams (e.g., “East Godavari” and “Godavari East”).
3. Jaro-Winkler distance ≤ 0.15 , for respellings and vowel swaps with 0.85 degrees of similarity.
4. Full Damerau-Levenshtein distance ≤ 2 , for ≤ 2 insertions, deletions, substitutions, and transpositions.